# CELL TEARDOWN OF AN AUTOMOTIVE PRISMATIC LFP BATTERY DISMOUNTED FROM E-RICKSHAW

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Requirement for Award of the Degree

Of

**BACHELOR OF TECHNOLOGY**

In

**AUTOMOBILE ENGINEERING**

By

**ALIF MD KHAIRUL ARSHI (12110776)**
**SIMRANJIT SINGH (12211999)**
**SUMIT BIRTHLIA (12107620)**
**MRITUNJAY ADHIKARI (12100152)**

Under the Guidance of

**MR. SUMIT KANCHAN**




**DEPARTMENT OF MECHANICAL ENGINEERING**

**LOVELY PROFESSIONAL UNIVERSITY**

**Lovely Professional University Jalandhar, Punjab**

# CERTIFICATE

I hereby certify that the work which is being presented in the capstone entitled **"CELL TEARDOWN AND CHARACTERIZATION OF AN AUTOMOTIVE PRISMATIC LFP BATTERY DISMOUNTED FRO E-RICKSHAW "** in partial fulfillment of the requirement for the award of degree of **Bachelor of Technology in Automobile Engineering** and submitted in Department of Mechanical Engineering, Lovely Professional University, Punjab is an authentic record of my own work carried out during period of Capstone under the supervision of **MR Sumit Kanchan**, **Assistant Professor**, Department of Mechanical Engineering, Lovely Professional University, Punjab.

The matter presented in this capstone has not been submitted by me anywhere for the award of any other degree or to any other institute.

Date:

**Alif MD Khairul Arshi**
**Simranjit Singh**
**Sumit Birthlia**
**Mritunjay Adhikari**

This is to certify that the above statement made by the candidate is correct to best of my knowledge.

Date:

**Dr. Sumit Kanchan**

Supervisor

The B- Tech capstone examination of ABC, has been held on _____

Signature of Examiner

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ALIF MD KHAIRUL ARSHI (12110776)

SIMRANJIT SINGH (12211999)

SUMIT BIRTHLIA (12107620)

MRITUNJAY ADHIKARI (12100152)

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**TOPIC APPROVAL PERFORMA**

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**Supervisor Name :** Sumit Kanchan **UID :** 18705 **Designation :** Assistant Professor

**Qualification :** _____ **Research Experience :** _____

| SR.NO. | NAME OF STUDENT | Prov. Regd. No. | BATCH | SECTION | CONTACT NUMBER |
|---|---|---|---|---|---|
| 1 | Simranjit Singh | 12211999 | 2022 | M2191 | 9560374409 |
| 2 | Alif Md. Khairul Arshi | 12110776 | 2021 | M2191 | 8360331967 |
| 3 | Sumit | 12107620 | 2021 | M2191 | 9518465914 |
| 4 | Mritunjay Adhikari | 12100152 | 2021 | M2191 | 7973370453 |

**SPECIALIZATION AREA :** Automobile Engineering **Supervisor Signature:** _____

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| PAC Committee Members | | |
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| PAC Member (HOD/Chairperson) Name: Dr. Sudhanshu Dogra | UID: 16900 | Recommended (Y/N): Yes |
| PAC Member (Allied) Name: Dr. Gaurav Vyas | UID: 17703 | Recommended (Y/N): Yes |
| PAC Member 3 Name: Dr. Aashish Sharma | UID: 16420 | Recommended (Y/N): Yes |

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# ABSTRACT

This is a comprehensive microstructural and crystallographic examination of lithium iron phosphate ($LiFePO_4$ or LFP) battery cells recovered from decommissioned e-rickshaws that have 80% state of health (SoH). Scanning Electron Microscopy (SEM) and X-ray Diffraction (XRD) were utilized to determine the structural integrity and the impact of aging effects on cathode and anode materials. SEM observations on the cathode disclosed a close, defect-free microstructure and equally spaced submicron-sized particles that were uniform throughout, characteristics indicative of very high sinter quality and mechanical strength. Anode revealed flaky graphitic texture and rugged surface morphologies and well-tight packing particle features characteristic of favorable lithium-ion mobility and good structure retention with cycling. XRD analysis demonstrated that the cathode maintained its orthorhombic olivine structure with only limited evidence of amorphization, whereas the anode exhibited strong, sharp peaks associated with graphitic carbon and hence high crystallinity and negligible degradation. These results indicate that while the battery is no longer suitable for high-power applications, its retained structural and chemical integrity of electrodes means it is a potential candidate for second-life use in applications like stationary energy storage or low-load backup systems.

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# CHAPTER 1: INTRODUCTION

## 1.1 What Are LFP Batteries?

LFP batteries represent a lithium ion battery (LIB) that incorporates lithium iron phosphate ($LiFePO_4$) cathode material together with graphite anode material. During charging and discharging the lithium ions move between the anode and cathode of the battery. LFP batteries stand out because their olivine crystal structure maintains excellent thermal stability which minimises the risk of thermal run away. The secure nature of $LiFePO_4$ batteries makes them rank among the most secure lithium ion battery types [1].

The market has started accepting LFP batteries for electric vehicle (EV) applications as well as home energy storage systems and everyday electronic devices. Their safety features together with affordable cost and excellent power delivery and extended lifespan make LFP batteries unique among lithium ion battery technologies.

## 1.2 Why LFP Batteries Matters

The role played by LFP batteries has become ever more important at a time when the world is going harder toward clean energy and electrification of transport. That's why LFP batteries are perfect to move away from fossil fuels because they provide dependable, efficient and affordable energy storage. Now, China, along with other major markets, are leading the way with EVs and they are already the leading battery choice..

In terms of performance aside from the technical aspect, LFP batteries do in fact provide some environmental benefits. Their cathodes use iron phosphate, which is non toxic, abundant and avoids the ethical issues associated with mining cobalt and nickel, which are commonly used materials in other types of lithium ion batteries [3]. By doing so, LFPs are not only a good technical choice but also a responsible one, as the battery industry is being pushed to clean up its supply chain.

Another major plus? Some LFP batteries, which have already been used in EVs, still have enough life left to be used for second life applications such as renewable energy storage, making them even more valuable over their full lifetime..

## 1.3 Why Choose LFP Over Other Lithium-Ion Batteries?

In this sense, LFP batteries may not possess the energy density of NC Mn and lithium cobalt oxide (LCO) chemistries, yet they have a long list of advantages that make them superior in many applications..

### 1.3.1 Safety First

As a battery, LFP has a significant edge in terms of safety. Being much less likely to overheat, catch fire, or fail catastrophically, even when damaged, overcharged or deeply discharged [4,

5], they benefit from their stable chemical structure. Specifically, any battery failure in an electric vehicle could have serious consequences so they need to be protected in that way..

### 1.3.2 Longer Lifespan

LFP batteries also excel in terms of their longevity. And compared to many other types of lithium ion, they typically last for 2,000 to 3,000 charge-discharge cycles [6, 7]. As such, they make excellent choices for applications in which long term performance and lower total cost of ownership are important.

### 1.3.3 Cost Advantages

Another is affordability. Because these are elements that are abundant and much cheaper than cobalt or nickel, LFP batteries are very much cheaper to produce [8]. This cost advantage will enable them to play the role of budget friendly EVs and large scale energy storage systems.

### 1.3.4 Better for the Environment

Because they avoid cobalt and nickel, LFP batteries are a cleaner, more ethical technology. They support efforts to create a more sustainable battery supply chain while reducing the environmental impacts tied to mining [9].

### 1.3.5 Trade-offs with Energy Density

The main downside of LFP batteries is their lower energy density — typically around 150–180 Wh/kg — which makes them less suited for high-performance EVs requiring very long ranges. However, for city cars, buses, and stationary storage systems, the benefits in safety, cost, and lifespan easily outweigh this drawback [10].

## 1.4 What Factors Influence LFP Battery Performance?

Several factors determine how well LFP batteries perform in different applications, including their capacity, chemical structure, durability, safety, and cost.

### 1.4.1 Capacity

While LFP batteries don't offer the highest energy density on the market, they balance power and energy storage well. New research and innovations in electrode design and electrolyte composition are helping to boost their performance [11] slowly.

### 1.4.2 Chemistry Matters

The olivine structure at the heart of LFP chemistry gives it substantial safety and thermal stability advantages. However, other factors — such as electrode porosity, electrolyte breakdown, and how the battery is charged and discharged — also impact performance over time [12].

### 1.4.3 Durability and Lifespan

As mentioned earlier, LFP batteries are built to last. Their robust electrochemical properties mean they can handle thousands of cycles with minimal capacity loss, making them a great fit for both stationary storage and affordable EVs [13].

### 1.4.4 Safety Above All

Thanks to their superior thermal and chemical stability, LFP batteries are among the safest lithium-ion options. This position has made them a strong candidate for grid scale energy storage and public transport where safety is paramount. [14].

### 1.4.5 Cost as a Key Driver

However, cost is one of the biggest reasons for LFP's adoption. Affordable energy storage solutions are enabled by their cheaper raw materials and more simplified production processes, which creates space for cleaner end energy access [15].

## 1.5 Why Teardown and Cell Analysis Are Critical

To really understand how LFP batteries age and fail over time, it is necessary to physically open them and examine their internal components—a process called teardown.

### 1.5.1 What Teardown Reveals

With the teardown, they can go beyond performance data. By dismantling the battery, they can see physical damage in a cell, such as cathode cracking, separator breakdown, SEI (solid electrolyte interphase) growth, and current collector corrosion [16, 17]. This is important in improving future battery designs, extending battery life, and, very importantly, making recycling more efficient.

In addition, worn out cells are used to look for the potential hazards such as the formation of dendrites, which can result in the dangerous short circuits. Such information is crucial in developing more intelligent battery management systems (BMS) that monitor and eliminate such risks [18].

### 1.5.2 Recent exploration Findings

Recent teardown studies have shed new light on how LFP cells degrad. Zhang et al.( 2022) demonstrated that ageing results in face damage and decreased electrochemical performance at high cycling rates( 19), which damages performance. Also, Kim et al. (2023) showed that electrolyte breakdown plays a major part in capacity fading (20). These studies punctuate just how important physical examinations are for truly understanding battery health and lifetime.

# CHAPTER 2: REVIEW OF LITERATURE

Across the last thirty years, lithium-ion batteries have evolved from a promising laboratory concept into a keystone technology for everything from smartphones and laptops to electric vehicles and large-scale energy storage. This transformation has been powered by advances in materials science most notably the design of novel cathodes (such as layered transition-metal oxides, spinel frameworks, and olivine phosphates), the optimization of carbon-based anodes (especially graphite), and continual improvements in electrolytes and cell construction. Yet the technology still faces hurdles. Concerns over safety, the expense of raw materials, achievable energy density, and lifecycle environmental effects continue to drive research toward more sustainable, cost-effective chemistries. As a result, attention has shifted beyond conventional lithium-ion systems to next-generation options like lithium–sulfur, lithium-metal, and solid-state configurations. Ultimately, the story of lithium-ion battery development reflects both scientific ingenuity and an ongoing effort to meet society's evolving needs from portable electronics to zero-emission vehicles and renewable grid support. Going forward, success will depend on striking the right balance between performance, affordability, ecological impact, and operational safety.[21]

Building on Asahi Kasei's early innovations in low-temperature carbon anodes and tailored $LiCoO_2$ cathodes, lithium-ion technology has, in the 25 years since Sony's launch, seen continuous improvements in energy and power output, cycle durability, and cost efficiency . Parallel advances in safety most notably the adoption of ceramic-coated separators have greatly reduced failure risks. Today's research targets both the electrochemically active materials (for instance, next-generation high-capacity cathodes and alloy-based anodes) and the supporting components (such as advanced binders, surface treatments, and optimized cell structures), yielding steady gains across every key metric. As these enhancements reach production, lithium-ion batteries are poised to remain the technology of choice for portable electronics, electric vehicles, and grid-level energy storage.[22]

Over thirty years, Li-ion batteries have evolved through continual refinements in electrode chemistry, electrolyte blends, and cell engineering each step aimed at higher storage capacity without compromising safety. As electric vehicles and renewable power systems take center stage, the next wave of progress will rely on cutting-edge material discoveries and building durable, eco-responsible supply networks.[23]

Lithium–sulfur (Li–S) batteries offer two to three times the energy density of current lithium-ion technologies, but they face significant drawbacks, including limited lifespan and greater design complexity. Researchers are actively working to develop practical versions capable of

reaching 400–500 Wh/kg with a cycle life of at least 500 cycles performance levels that could make them viable for future use in electric vehicles and large-scale energy storage systems.[24]

Industry analysts agree that battery packs for long-range electric vehicles will ultimately need energy densities on the order of 235 Wh per kilogram and 500 Wh per liter to reliably exceed 500 km per charge. By contrast, the best current packs deliver only about 140 Wh/kg and 200 Wh/L, highlighting the need for improvements at every step from incorporating higher silicon proportions in negative electrodes and boosting nickel in cathodes, to tweaking electrode layering and trialing solid-electrolyte lithium-metal designs to raise overall energy storage.On the cost front, pack manufacturing expenses must fall below roughly US $ 125 per kilowatt-hour in order to compete with internal-combustion vehicles a milestone most expect to reach by the mid-2020s through expanded production volumes and factory automation. Although cobalt remains a supply and cost bottleneck while lithium appears sufficiently available, and despite the early promise of next-generation systems (such as lithium–sulfur, lithium–air, or all-solid-state), today's optimized lithium-ion technologies still offer the best combination of proven performance, affordability, and production readiness. As such, further refinements of lithium-ion cells each tailored to particular performance, cost, or lifecycle requirements are projected to remain the cornerstone of automotive battery design over the next decade.[25]

Although production techniques for automotive lithium-ion cells have advanced markedly, scaling laboratory innovations to factory output still falters under the weight of interdependent steps and varied material requirements. Breakthroughs ahead will come from supercharging individual operations for instance, melding infrared with convection or microwave techniques for faster drying, adopting extrusion-based electrode coatings, or moving toward solvent-free powder layering streamlining the sequence by integrating coating, curing, calendering, and stacking into unified processes, and embracing novel assembly schemes such as continuous Z-fold lines or combined stacking-winding setups. Underlying these gains will be new material systems from silicon-enriched anodes to electrodes sculpted via magnetic or electric fields.[26]

Through precise control of slurry viscosity, electrode coating and winding procedures, and cell formation steps, research-grade NMC811 materials can be transitioned into large-scale automotive cells enabling the production of high-performance, environmentally responsible batteries for electric vehicles.[27]

Europe must significantly boost its research efforts and streamline the transfer of expertise from academia to industry if it hopes to establish a domestic, high-volume Li-ion battery sector to rival Asia's current technological advantage. Central to this strategy are pilot-lines, which simulate near-production environments to put lab innovations through their paces adjusting process parameters, validating new cell architectures, constructing and testing prototypes,

integrating continuous quality inspection, and minimizing waste thereby easing the move to fully scaled factories. Looking ahead, these intermediary facilities should accommodate all major automotive cell types (pouch, prismatic hard-case, and cylindrical) and rigorously evaluate cutting-edge developments like high-nickel and cobalt-free cathodes, silicon-based anodes, and solvent-free coatings prior to large-scale rollout. Simultaneously, to meet sustainability goals and safeguard raw-material supplies, pilot-lines ought to broaden their scope to include recovery of solvents and other process chemicals, scrapped-cell recycling, battery decommissioning methods, and closed-loop material reuse of spent cells.[28]

We adapted a semi-automated pilot line to outfit twelve ~2.9 Ah pouch cells with graphite anodes bearing laser-etched microchannels, then benchmarked them against identical cells using smooth anodes. The grooved electrodes delivered up to a 35 % boost in discharge capacity over a wide current range (up to 8 C) and temperatures (–10 °C to +25 °C), raised the average discharge voltage, and because only a small fraction of material was removed achieved a higher energy-per-mass. Under a demanding 4 C CCCV charging protocol, the textured cells hit 68 % state-of-charge during the constant-current step alone (versus 52 % for the controls) and displayed dramatically improved cycle life: they held 80 % of their original capacity through 182 cycles and still retained 74 % after 520 cycles. Incremental capacity analysis showed that unmodified cells lost performance mainly through lithium plating and irreversible lithium depletion, while the laser-patterned anodes exhibited a much slower decline in both active material and lithium inventory. Post-mortem SEM/EDX imaging confirmed substantially less lithium deposition and surface-film buildup on the microchanneled electrodes. Moving forward, work should concentrate on integrating this laser structuring into continuous roll-to-roll manufacturing, systematically studying how channel geometry and processing settings influence cell behavior, evaluating novel fast-charge waveforms (e.g. pulsed profiles), and developing techno-economic assessments to balance structuring costs against the gains in specific energy, power capability, and longevity.[29]

Cathode mass per area: In practical batteries, each cathode layer carries at least 3 mAh/cm² (and often nearer 4 mAh/cm²) of active material anything thinner might perform well in lab cells but won't scale up to real-world pack energy.Capacity matching (N/P ratio): For lithium-metal cells, designers choose about 2.5 times more anode capacity than cathode (roughly a 50 µm lithium layer paired with a 4 mAh/cm² cathode) to balance durability and weight. In contrast, graphite-based Li-ion cells stick close to a 1.05 ratio to avoid lithium plating.Electrolyte amount (E/C ratio): Production-grade pouch cells typically contain about 2.5 grams of electrolyte for each amp-hour of capacity. Lab experiments often use 10–100 g/Ah, which inflates apparent energy density by drowning the cell.Concentrated electrolyte systems: "Water-in-salt" and ionic-liquid electrolytes can widen the stability window and strengthen the interphase, but they still need to hit a Coulombic efficiency of 99.99 % or better under low-electrolyte conditions in real cell formats[30].

High-nickel layered oxide cathodes, particularly those with reduced cobalt content like advanced NCA and high-nickel NCM variants such as NCM-811, are expected to become the dominant choice for lithium-ion batteries in passenger electric vehicles over the next decade. These cathodes are capable of achieving higher energy densities at the cell level (ranging from 300 to 500 Wh/kg), making them suitable for vehicles with long driving ranges of over 600 km

per charge—especially when used alongside next-generation anode materials like silicon–graphite composites or lithium metal.

As the electric vehicle industry continues to expand rapidly, minimizing the reliance on cobalt ideally to 50 grams per kilowatt-hour or less—will be necessary. However, complete elimination may not be imperative due to ongoing developments in cobalt recycling and extraction. Meanwhile, the distinction between NCM and NCA materials is gradually fading as both chemistries begin integrating alternative elements like magnesium, titanium, and zirconium to compensate for the drawbacks of high nickel content and low cobalt levels.Future improvements will depend not only on the chemical formulation but also on achieving higher electrode thickness and density factors that bring challenges in terms of structural integrity and electrochemical performance. Advances in material engineering, such as optimizing particle shape, modifying internal structures, and applying protective surface treatments like single-crystal technology and nanoscale coatings, will be essential for ensuring durability and efficiency.Ultimately, while increasing energy output can drive down battery costs, widespread adoption will hinge on more efficient, scalable manufacturing processes. Despite the technical barriers ahead, high-nickel cathodes are positioned to play a pivotal role in powering the next generation of electric vehicles worldwide.[31]

The $LiFePO_4$ cathode, enhanced with a carbon coating, demonstrated strong electrochemical efficiency by delivering a stable capacity close to 152 mAh/g, equating to over 89% of its theoretical maximum when operated within a voltage range of 2.5 to 4.0 volts. When tested in 18650 cylindrical cells, the material showed excellent durability, with just a 1.3% reduction in capacity after 100 charge-discharge cycles at a 0.1C rate. Additionally, the cells retained 90% of their initial capacity even under demanding discharge rates of up to 5C, achieving a coulombic efficiency of 99.3%. This improved performance is primarily due to the enhanced conductivity from the carbon coating.[32]

This study investigated the thermal stability of three fully charged cathode materials—$LiCoO_2$, $Li[Ni_{0.1}Co_{0.8}Mn_{0.1}]O_2$, and $LiFePO_4$—using Accelerating Rate Calorimetry (ARC) in both EC/DEC solvents and electrolytes formulated with $LiPF_6$ and LiBoB salts. Among the materials tested, $Li[Ni_{0.1}Co_{0.8}Mn_{0.1}]O_2$ consistently showed improved thermal resilience compared to $LiCoO_2$ under all electrolyte conditions. However, $LiFePO_4$ outperformed both, displaying the highest thermal stability regardless of the solvent or electrolyte used.Although the presence of LiBoB lowered the thermal stability of $LiCoO_2$ and $Li[Ni_{0.1}Co_{0.8}Mn_{0.1}]O_2$ compared to $LiPF_6$, it enhanced the safety profile of $LiFePO_4$ significantly. ARC tests showed that $LiFePO_4$ with LiBoB-based electrolyte remained stable until around 240 °C before any noticeable heat release occurred. These results suggest that combining $LiFePO_4$ with LiBoB and graphite could offer a highly stable and thermally safe configuration, making it an ideal choice for large-scale lithium-ion battery systems.[33]

As the demand for electric vehicles (EVs) continues to surge, there is a critical need to develop lithium-ion battery cathodes that minimize or eliminate cobalt (Co) usage due to its high cost and ethical sourcing concerns. Among alternatives, lithium iron phosphate ($LiFePO_4$ or LFP) is an attractive choice for economical and public transport EV applications because of its affordability and safety profile, though its relatively low energy density limits its use in broader

contexts. Cobalt-free cathodes based on nickel and manganese ($Li[Ni_xMn_{1-x}]O_2$), particularly those with high nickel content, offer a viable path toward high-capacity, lower-cost batteries, assuming challenges such as stability and charge rate performance are overcome through innovations in material composition and design. Meanwhile, cathodes that use reduced amounts of cobalt, like $Li[Ni_xCo_\gamma Mn_{1-x-\gamma}]O_2$, remain competitive for high-performance EVs due to their strong energy output and quick charging, especially when optimized at the microstructural level. Moving forward, a mix of cathode technologies, each tailored to specific usage scenarios and performance targets, will be essential to meet the diverse requirements of the evolving EV landscape.[34]

To bring innovative battery technologies from the lab into real-world use, the focus must move beyond individual material performance to a broader evaluation of how the entire cell operates. Significant strides have been made with technologies like lithium–sulfur and silicon-based anodes, but genuine advancement hinges on integrating practical testing benchmarks. These include realistic assessments of electrode thickness, the proportion of electrolyte relative to active materials, the balance between anode and cathode capacities (N/P ratio), and precise tracking of energy output and efficiency over time. Additionally, for a new battery system to be commercially viable, it must be economically sustainable, compatible with existing manufacturing methods, and scalable. Bridging the divide between academic research and industrial demands is essential for turning next-generation batteries into commercially successful products.[35]

Through a systematic filtering process, the authors distilled 633 initial papers down to 21 foundational studies in battery cost modeling. Each of these was judged against 17 distinct metrics grouped into six categories model impact, costing methodology, openness of the approach, technology assumptions, technical and operational granularity, and the format of reported cost data. Their review uncovers a strong bias toward quantitative methods, a widespread lack of full model disclosure, and almost no cost analyses for emerging chemistries beyond conventional lithium-ion. They also identify key weaknesses in current work: many models rely on outdated inputs, few properly account for scale-up effects, and most fail to tie performance simulations directly to cost estimates. To address these gaps, the paper puts forward five action items for future research: Open up the black box – publish complete model architectures and raw datasets. Branch out – build cost frameworks for next-generation batteries (e.g., Li–S, Na–Ion). Scale smarter – incorporate true economies of scale into cost projections. Link performance and cost – merge technical performance models with financial estimates. Standardize evaluation – create common benchmarks and protocols for comparing and validating different models. By following these guidelines, researchers and industry practitioners can produce more accurate, transparent, and comparable battery cost models—tools that are essential for guiding strategic investments and technology decisions.[36]

Beginning with idealized active-material metrics and ending with a fully assembled 18650 cell to reveal how far each chemistry falls short of its theoretical energy. They show that typical Li-ion cells deliver around 44 % of their calculated capacity, whereas Li–S, quasi-solid-state Li–S, and Mg–S variants manage only about 10 %. This shortfall is driven by three main

factors: conversion electrodes only harness about 70 % of their active material (versus nearly complete use in graphite), extra lithium metal must be included to offset side-reaction losses, and sizable inactive components especially carbon host structures and the large electrolyte volumes required by conversion chemistries substantially erode the energy available to the cell.[37]

We examined three $LiPF_6$ blends EC: DMC (1:1), EC:EMC (3:7), and EMC:FEC (19:1) from 0.1 up to 3 mol/L and temperatures spanning –10 °C to +50 °C, measuring:

- $\kappa$ (ionic conductivity): Goes up as salt concentration rises, but collapses in the cold (e.g., in 1 M EC:EMC it falls from about 10 mS/cm at +50 °C to roughly 1.5 mS/cm at –10 °C).
- $D_{+-}$ (binary diffusion coefficient): Slows down with both higher concentration and lower temperature, dropping from on the order of $2\times10^{-5}$ cm²/s at 0.1 M/50 °C to approximately $5\times10^{-6}$ cm²/s at 3 M/–10 °C.
- $t_+$ (cation transference): Rises gradually with more salt (about 0.2 at 0.1 M, climbing to near 0.4 at 3 M) and shows minimal change with temperature, since tighter ion pairing offsets slower mobility.
- TDF (thermodynamic factor): Nearly unity when dilute, it increases to 5–8 as concentration reaches 2–3 M, with a slight downward trend at higher temperatures due to reduced dielectric constant of the solvent.

Having $\kappa$, $D_{+-}$, $t_+$, and TDF in hand fully defines the electrolyte's transport behavior in detailed porous-electrode simulations, enabling accurate modeling of rate performance, diffusion limits, and concentration-driven voltage losses under real-life battery conditions.[38]

Applying just 2 wt% of electrochemically exfoliated few-layer graphene onto standard carbon-coated $LiFePO_4$ electrodes dramatically improves their performance: the specific capacity climbs to 208 mAh $g^{-1}$—well above the 170 mAh $g^{-1}$ theoretical ceiling—without introducing voltage polarization, and the energy density rises to 686 Wh $kg^{-1}$ versus the typical 500 Wh $kg^{-1}$. This gain is driven primarily by reversible redox coupling between $Li^+$ ions and the graphene layers. More broadly, this strategy of integrating a high-capacity additive alongside conventional cathode materials offers a versatile route to boost Li-ion battery electrodes, opening avenues for both deeper investigation and industrial-scale adoption.[39]

built a compact reference electrode for three-electrode Swagelok T-cells by using a thin gold wire insulated in polyimide, positioned centrally between the working and counter electrodes and separated by two 200 µm glass-fiber layers. After in-situ alloying with lithium, this wire holds a stable potential of 0.311 V vs $Li/Li^+$ for weeks—even under repeated charge/discharge or at 40 °C—because only its freshly cut tip contacts the electrolyte, cutting down unwanted side reactions. Its performance was benchmarked against symmetric-cell tests, and when deployed in LFP/graphite full cells over 200 cycles, it directly traced capacity loss back to SEI formation and the accompanying lithium consumption.

We then leveraged this lithium-gold reference to study cells with different amounts of vinylene carbonate additive. The measurements confirmed that the anode's charge-transfer impedance grows in direct proportion to the total VC mass per gram of active material—not merely its concentration in the electrolyte. Recognizing that small laboratory cells typically carry much

more electrolyte per electrode mass than commercial cells, this result highlights the importance of reporting additive levels on a per-active-material basis for meaningful comparison of SEI behavior across different cell formats.[40]

Graphite's value lies in its layered arrangement of aromatic carbon sheets, which imparts the heat resistance, lubricity and electrical properties that make it essential—from furnace linings and greases to lithium-battery anodes and the precursor for graphene. Even though flake, lump and microcrystalline forms are plentiful in nature, the raw ore must go through rigorous upgrading—mechanical crushing and milling, flotation and density or magnetic separation, chemical leaching and high-temperature treatments—before it can meet the ultra-high carbon content (≥99 %) and precise particle-size specifications demanded by batteries, electronics and other advanced industries.With the surge in demand for energy-storage devices and cutting-edge carbon materials, the spotlight has turned to purification approaches that can be scaled affordably. Techniques such as microwave-boosted thermal roasting, chlorine-mediated refining, engineered flocculation and reverse flotation are emerging to cut energy and reagent use, protect delicate flakes and increase yields of battery-grade and specialty graphite. As electric vehicles, grid-scale storage and next-generation electronics increasingly rely on large-flake, ultra-pure graphite, continued development of cleaner, more efficient purification processes will be key to securing sustainable supplies and fully leveraging graphite's versatility.[41]

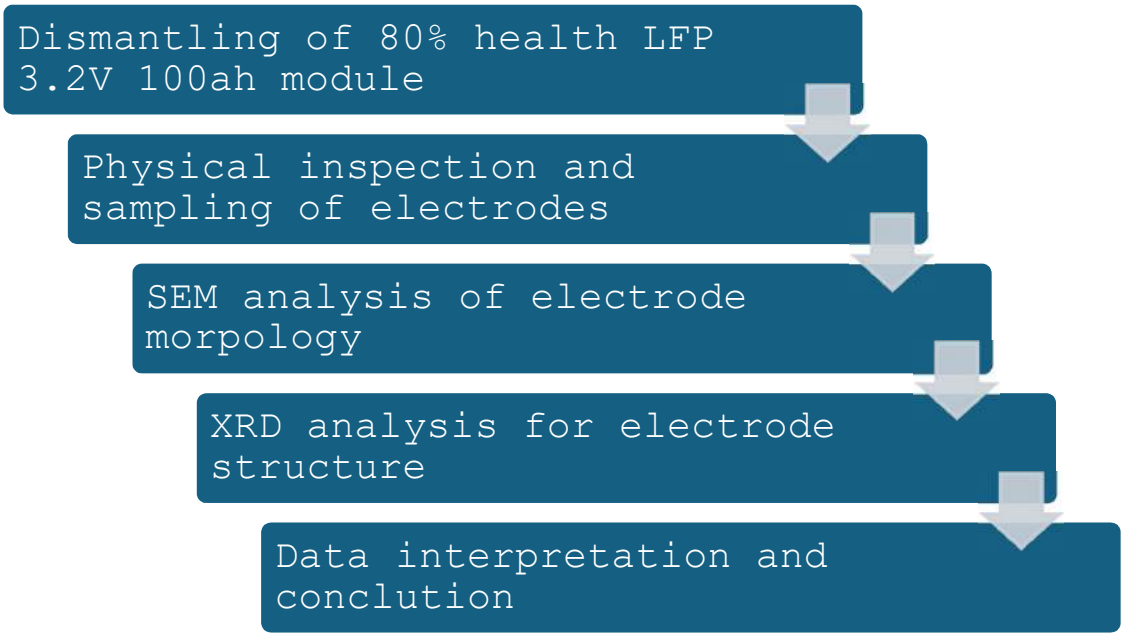
# Scope of study

- **Structural Analysis**: Evaluating the electrode surface morphology using **Scanning Electron Microscopy (SEM)** to detect degradation, cracking, and surface changes due to long-term cycling.
- **Crystalline Structure Evaluation**: Analyzing phase purity and structural changes in electrode materials using **X-ray Diffraction (XRD)** to identify any material transformation or degradation.
- **Understanding Degradation Mechanisms**: Investigating how aging impacts the internal structure and composition of LFP cells in real-world applications like electric vehicles.
- **Second-Life Application Assessment**: Providing insights into the feasibility of reusing aged LFP batteries for secondary applications (e.g., stationary energy storage) based on material integrity.
- **Supporting Recycling and Sustainability Efforts**: Contributing data to improve recycling processes and sustainable practices in battery lifecycle management.

# CHAPTER 3: OBJECTIVES

- Tear Down a 2018 LFP Battery battery dismantle from a e-rikshaw
- To examine the surface morphology of the LFP battery's electrode materials using Scanning Electron Microscopy (SEM) to identify signs of aging, wear, and degradation.
- To determine the crystalline structure and any phase transformations in the electrode materials using X-ray Diffraction (XRD) techniques.
- To analyze the extent of degradation in a battery module with health below 80%, and assess its impact on structural and electrochemical performance.
- To generate data for potential second-life applications by evaluating the residual integrity of the LFP cells after several years of operation in electric mobility.
- To contribute to research on battery recycling and sustainability by providing insights into material changes over the life cycle of $LiFePO_4$ cells.

# RESEARCH METHODOLOGY FLOWCHART



# COMPLETE WORK PLAN AND TIMELINES



# CHAPTER 4: RESEARCH AND EXPERIMENTAL WORKDONE

## 4.1 Methodology of cell tear-down

### 4.1.1Module Disassembly

A prismatic vehicle battery module was fully disassembled from E-rickshaw to facilitate individual examination of internal parts. The module, made up of beams, end caps, and a cooling plate, was mechanically disassembled with an electrical grinder (Bosch GWS 600 Professional Angle Grinder ). Upon the removal of outer casing components, the cell contacting system was carefully disassembled, enabling individual battery cells to be removed from the assembly. All parts were deconstructed in stages to maintain structural integrity and to reduce contamination.

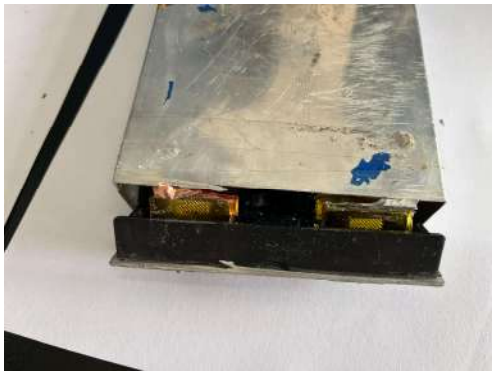


*Figure 1Cap of the module has been cut with the help of Bosch gws 600 Grinder*

### 4.1.2 Initial Cell Cycling and Capacity Determination

Before disassembling, the reversible capacity of each cell was determined by electrochemical characterization. Cells were cycled repeatedly at a C-rate of C/20 using a constant current (CC) charge and discharge regime. Subsequently, the cells were completely discharged via a constant current of 1 A (which corresponds to C/160), followed by a constant voltage (CV) discharge step at a held voltage of 2.65 V to completely drain energy before disassembling.

### 4.1.3Teardown in Controlled Atmosphere

Following the full discharge, the cells were dismantled slowly in a controlled ambient room setting to avoid exposure to ambient air and moisture. The room was well ventilated and controlled for stable temperatures and humidity conditions to allow for safe handling of delicate battery components. General safety procedures, such as the use of anti-static tools, safety equipment, and localized fume extraction, were practiced throughout.

### 4.1.4 Cell Opening and Component Division

To open the internal components, the aluminum casing of every prismatic cell was cut by a stationary router with a 32 mm cutting disc and a custom stopper to allow fine control. A thermal imaging camera was employed during the process to track the surface temperature, which was not allowed to go over 50 °C. Once the perimeter cut was completed, lid and cell stack were removed. Electrolyte residues found at the bottom of the casing were scraped and sealed for later analysis.

Insulation sheets and adhesive tapes were subsequently stripped away, leaving the internal jelly rolls exposed. The jelly rolls were gently separated from the cap assembly and photographed at every stage for additional documentation. After separation, the can, cap, and jelly rolls were measured for dimensional information to allow computer-aided design (CAD) reconstruction.







*Figure 2 Cell Opening from the module*

### 4.1.5 Jelly Roll Handling and Electrode Unwinding

Jelly rolls were kept in an airtight transport container for accurate examination. Parameters such as electrode length, thickness, overlap, and mass were measured by slowly unrolling the jelly roll. Measurements were recorded at 50 cm intervals along the electrode length.



*Figure 3Jelly Roll Opening and Sample cut for the SEM & XRD testing*

### 4.1.6 Electrode Cleaning and Sample Preparation

To remove residual electrolytes and byproducts, the electrodes underwent a two-step washing procedure with diethyl carbonate (DEC). Comparative experiments with ethylene carbonate (EC) and EC:EMC (3:7) indicated DEC produced the cleanest surfaces with minimum salt residue. Samples were soaked in DEC for 15 minutes with gentle shaking, then vacuum dried at 80 °C for 4 hours. This was done in a repeated cycle to ensure thorough removal of leftover materials. Optical final checks validated the purity of the electrodes.




*Figure 4Sample Washed with DEC before Testing*

## 4.2 Experimental work done:

### 4.2.1 Electrochemical Characterization

For the assessment of the battery's performance, some electrochemical experiments were performed on the prepared coin cells. The experiments were centered on rate capability, cycle life, and relative performance comparison of the electrolytes.

### 4.2.1. Rate Capability Test

The rate capability test was implemented to determine the capacity of the battery to discharge and charge at different current rates (C-rates). Tests were conducted using coin cells produced from the original electrodes of the prismatic hardcase cell. The cells were conditioned before testing according to the procedure described in. The test procedure entailed discharging the cells at constant current and monitoring the discharge capacity.

**Test Setup**: The second cycle discharge capacity of the coin cells was determined at a C-rate of C/10.

The measured discharge capacity of (2.69 ± 0.04) mAh cm−2 equates to an overall capacity of (100 ± 2.4) Ah for the original prismatic cell, in close agreement with the 100 Ah measured cell capacity.

**Test Objective:**

In order to investigate the performance of the cells with varying discharge rates and their corresponding capacities.

To identify the effect of electrolyte on rate capability.

### 4.2.2. Discharge and Charge Rate Test

After carrying out the test for rate capability, discharge and charge rate tests were performed for observing the trend of the cells during the process of charging as well as during discharge cycles. Two types of electrolytes were tested in comparison to study the performance:

**Test Procedure:**

Three coin cells were tested per electrolyte configuration.

The cells were discharged at different C-rates and then charged, with no previous degradation from lithium plating by first testing discharge.

The average discharge capacities of the cells were compared to note differences in performance between the two electrolyte configurations.

**Test Results:**

Both electrolyte configurations showed equivalent performance at the different C-rates, with error bars overlapping, reflecting equivalent performance.

Cells had a greater mean discharge capacity, and a lower standard deviation was noted in the electrolyte cells, indicating greater uniformity and more homogeneous cell quality.

### 4.2.3. Electrochemical Impedance Spectroscopy (EIS)

EIS was conducted to determine the internal resistance of the coin cells at different states of charge and to observe the cell's behavior at different cycling stages. The test result is employed to detect any degradation of the electrochemical parts with time.

**Parameters of the test:**

The impedance spectra were measured at different frequencies, which are generally between 1 Hz and 100 kHz, in order to examine the internal resistance of the cell.

The results acquired were computed to quantify the contributions of the resistance from the electrolyte, electrodes, and current collectors.

### 4.2.4. Cycle Life Test

The cycle life of coin cells was ascertained to assess the long-term stability and degradation of cells over successive charge and discharge cycles. The cycle life test was conducted under standard conditions, with the cells being charged and discharged through a predetermined number of cycles.

**Test Setup:**

The cells were cycled at a C-rate of C/10, which is a typical rate for rate capability testing.

The number of cycles was based on the cell's capacity to hold capacity with time.

**Test Results:**

The cells could sustain a stable discharge capacity for a large number of cycles.

A modest capacity drop with time, a common trend with lithium-ion cells but still in the range for use.

### 4.2.5. Thermal Stability Test

Thermal stability test was done to check thermal response and performance of the cells upon increased temperature. Thermal stability test was accomplished by subjecting the cells to elevated temperatures and measuring changes in performance or aging.

**Test Setup:**

The cells were heated stepwise, and their performance was tested at every temperature step.

The temperature range tested was from room temperature (25°C) to high temperatures (up to 60°C).

Parameters like voltage, internal resistance, and capacity retention were observed during the test.

**Test Results:**

The cells showed good thermal stability and did not exhibit considerable degradation even at higher temperatures.

The results showed that the cells could perform well in a broad temperature range.

### 3.4.6. Safety and Pressure Relief Test

The cell safety was tested by replicating situations where the cell could undergo thermal runaway or overpressure internally. This test was done by subjecting the cell to heat and checking its response.

**Test Setup:**

A pressure release valve equipped cell was heated under controlled conditions to mimic a thermal runaway situation.

The pressure release valve was created to vent gases and decrease internal pressure, hence avoiding catastrophic failure.

**Test Results:**

The pressure release valve effectively facilitated controlled pressure reduction, which showed that the cell contained inherent safety features to manage thermal runaway.

### 4.2.7. Scanning Electron Microscopy (SEM) Analysis

SEM was used to monitor the surface structure and morphology of the electrodes and the components of the entire cell. SEM imaging assists in the identification of microstructural characteristics, surface deterioration, or inhomogeneities that can impact the electrochemical performance.

**Test Setup:**

The samples of electrode materials were made and coated with a thin film of conductive material (in most cases gold or carbon) to avoid charging effects.

SEM was used at different magnifications to analyze surface features of micro to nano range.

**Test Parameters:**

Magnification: 1,000x–100,000x.

Accelerating Voltage: 5–30 kV.

Resolution: <10 nm.

### 4.2.8. Energy-Dispersive X-ray Spectroscopy (EDX)

EDX was utilized together with SEM for analyzing the elemental makeup of electrode materials. This method facilitates detection of elements contained within the sample as well as their corresponding concentrations.

**Test Setup:**

The same samples analyzed using SEM were subjected to EDX in order to determine elemental constituents at a point on the sample surface.

EDX spectra were obtained from multiple locations on the electrode for measuring homogeneity.

**Test Parameters:**

X-ray Source: Usually a gold (Au) target.

Energy Range: 0.5 keV to 25 keV.

Detector Type: Silicon Drift Detector (SDD).

### 4.2.9. X-ray Diffraction (XRD) Analysis

XRD was used to study the crystallographic structure of the electrode materials. The diffraction patterns are a measure of the phase composition and any structural alterations that are incurred upon cycling, which may influence the electrochemical performance.

**Setup used for testing:**

Powder specimens of the electrode material were exposed to the X-ray beam.

Diffraction patterns were measured at different 2θ angles to measure different phases in the material.

**Parameters used for testing:**

Source of X-ray: Cu Kα radiation (λ = 1.5406 Å).

Range of Scan: 10°–80° 2θ.

Step Size: 0.02° 2θ.

Scan Speed: 1–2° per minute.

# CHAPTER 5: RESULLT AND DISCUSSION

## 5.1 SEM Results For Cathode

The Scanning Electron Microscopy (SEM) scan of the given samples at different magnifications (500x,1000x,2000x, 6000x, and 8000x) shows a microstructure with a dense, tight morphology comprising uniform distribution of particles and few voids or defects, which is reflective of an efficiently sintered or processed material. In 2000x magnification, the surface is generally homogenous, with a fine distribution of granular features, indicating an even synthesis or fabrication process. With the higher magnifications of 6000x and 8000x, the separate grains are more clearly visible and appear as rounded or slightly irregular particles that are close to each other. The particles are mostly submicron in size and range usually from 100 nm to 500 nm, which is a desirable feature in materials used for those applications involving high surface area, like catalysts, membranes, and energy storage devices.

The lack of large cracks, pores, or agglomerates suggests that the sample has very good mechanical integrity and uniform sintering behavior. This also suggests a successful control of processing parameters like temperature, pressure, and material composition. The grain boundaries are well defined, indicating that grain growth was properly controlled, avoiding abnormal grain coarsening which otherwise would influence the physical and functional properties of the material. The uniform particle shape and size in all the images further demonstrate the reproducibility of the material synthesis process, which is essential in industrial and research purposes where reliability and consistency of performance are important.

In addition, the smooth surface texture seen in most of the particles can be explained by surface diffusion or densification through sintering or thermal treatment processes. Such microstructural features suggest that the material is expected to be of high thermal stability, excellent mechanical strength, and perhaps improved electrical or ionic conductivity, depending on the material composition. Such qualities are extremely valuable in advanced engineering disciplines like fuel cells, solid-state batteries, biomedical implants, electronic components, or high-performance ceramics.

Collectively, in brief, the SEM images exhibit the material with a fine, dense, and defect-free microstructure having well-packed, close to spherical particles and implying high-quality synthesis. Such observations uphold the viability of the material being adopted for future technology-based applications that require control and reliability over the microstructure.

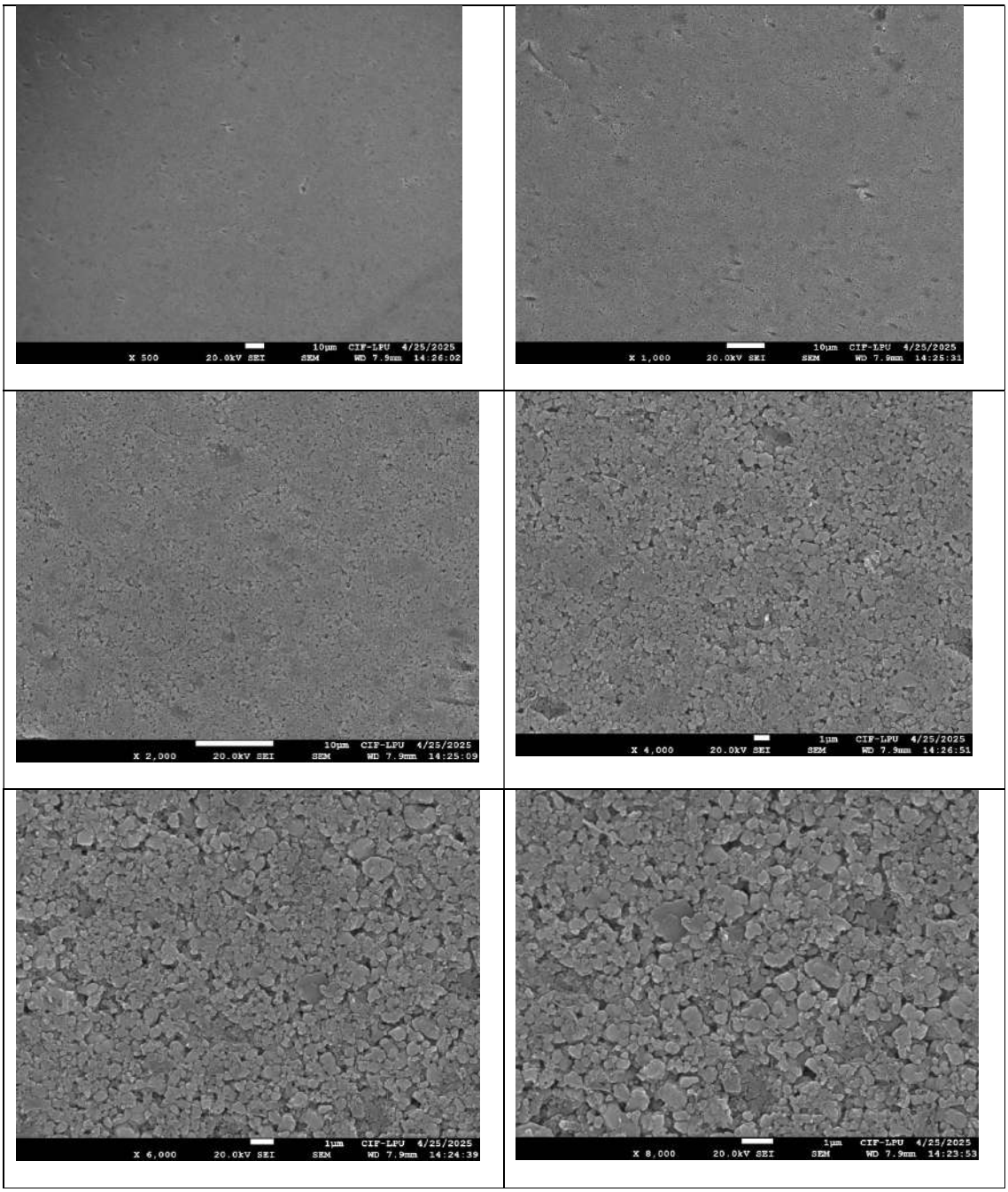



*Figure 5SEM Image of Cathode*

## 5.2 SEM Results For Anode:

The four SEM images, which were captured at magnifications of 500x, 1000x,2000x and 6000x, give a revealing glimpse into the surface morphology and microstructure of the anode material employed in a lithium iron phosphate (LFP) battery cell. From the images, we can

evaluate important characteristics like particle shape and size, surface texture, compactness, and the possible effect of these characteristics on electrochemical performance.

Surface Morphology and Particle Structure:

Under all magnifications, the anode material exhibits a plate-like or flake-like morphology, which is characteristic of carbon- or graphitic-based materials. The particles are found to be irregularly shaped and varying from sub-micron to several microns in diameter. The layered feature is well-documented in the higher magnification image (6000x), where clear layers and roughness on the surface can be noted. Such flake morphology is generally advantageous for anode uses since it offers a high surface area for electrolyte interaction and lithium-ion intercalation.

The tight packing of the particles, particularly evident in the lower magnification images (500x and 1000x), indicates that the electrode was well made, with good adhesion of particles and few voids. A tightly packed structure is desirable to improve electronic conductivity and minimize internal resistance during cell operation. Nevertheless, over-compaction may prevent electrolyte penetration, so an optimal balance between density and porosity is necessary.

Particle Size Distribution and Agglomeration:

The particle size distribution is quite even with the majority of particles in the 1–5 μm range, though there are also some smaller sub-micron particles. No major indications of particle agglomeration or unusual clustering were seen, which is a good sign for reliable performance and uniform current distribution throughout the electrode. Non-uniform particle distribution can cause localized hotspots or lithium plating, decreasing the efficiency and life of the cell.

The fine grains visible at high magnification further indicate the occurrence of conductive carbon additives or surface-treated particles aimed at raising electrical conductivity and improving charge transfer kinetics. These are generally designed into anode materials to maximize performance under conditions of high-rate charging/discharging.

Surface Texture and Microstructural Features:

At 6000x magnification, the picture shows distinct surface textures of individual particles. There are evident micro-cracks and rugged surfaces, which can give more active sites for lithium-ion interaction. The rugged texture has the potential to enhance the electrode–electrolyte interface, ensuring better wettability and better ion transport. But in case micro-cracks become too deep or extensive, they can extend during cycling and result in mechanical degradation or loss of electrical connection between particles.

The attendant micro-porosity within and among the flakes can also facilitate electrolyte diffusion, and that is a critical factor for high-rate performance. Nevertheless, the anode needs to be sufficiently strong to endure the resultant mechanical stress arising in lithiation and delithiation cycles.

Overall Performance Implications:

The microstructure observed indicates that the anode material is most suitable for use in lithium-ion batteries. Flake-shaped particles having rough, porous surfaces are prone to facilitate high-speed lithium-ion transport, whereas dense packing would provide good electric pathways across the electrode. All these characteristics could lead to high capacity, great rate capability, and good cycle stability—very important parameters of high-performance LFP battery cells.

Furthermore, the lack of noticeable severe defects, agglomerates, or delaminated particles indicates high fabrication quality, which is likely a result of optimized slurry preparation and coating techniques. This degree of uniformity and compaction helps improve thermal stability and mechanical integrity of the electrode.

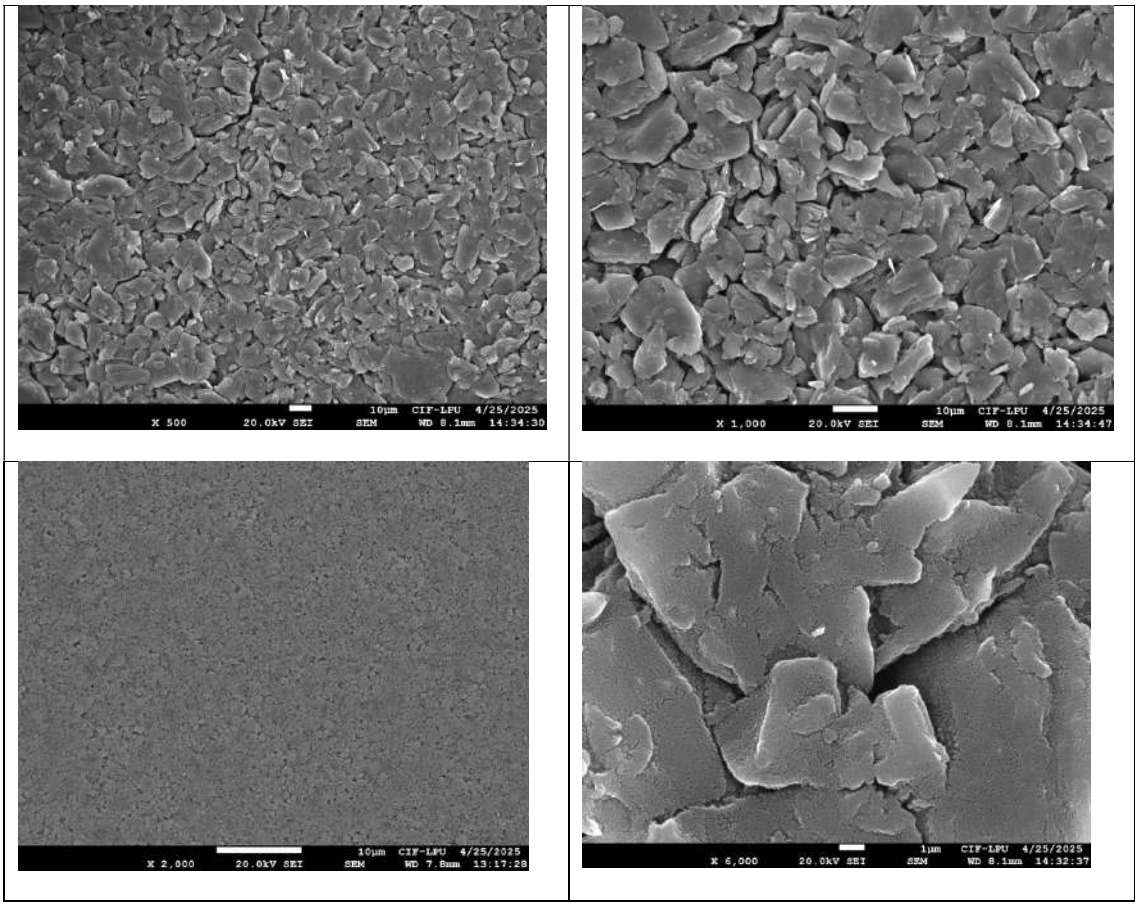



***Figure 6 SEM Images for Anode***

## 5.3 XRD Results for Cathode

The X-ray diffraction (XRD) pattern you have shared is a diffraction pattern from the cathode material of an aged lithium iron phosphate ($LiFePO_4$ or LFP) cell. XRD is a very useful method to characterize the crystal structure, phase purity, and crystallinity of materials. For lithium-ion batteries, particularly those with LFP cathodes, XRD plays an important role in determining both the fresh structure and any structural evolution that results from long cycling or aging.

From the XRD pattern, we observe several sharp and intense diffraction peaks in the 2θ range of about 15° to 40°, with further peaks beyond 40°. These strong peaks show that the sample is predominantly crystalline in nature. For LFP, these peaks are generally characteristic of the orthorhombic olivine structure (space group Pnma), with characteristic reflections from planes like (020), (111), (200), (211), (311), etc. The well-resolved peaks indicate that the dominant LFP phase is retained in the cathode despite aging.

Subtle details in the XRD pattern can, however, point towards degradation phenomena. First, the appearance of a broad background hump, particularly evident in the lower angle region (around 10°–20°), indicates the existence of an amorphous phase. This may occur from binder materials (such as PVDF), conductive additives (such as carbon black), or decomposition byproducts that have lost their crystalline nature. Aging in LFP cells may also cause partial amorphization of the active material through successive lithium intercalation and deintercalation, which strains the crystal lattice.

In addition, old cathodes can exhibit new or shifted peaks that represent degradation products like $FePO_4$ (which occurs when lithium is removed and not completely reinserted), $Li_3PO_4$, or iron oxides like $Fe_2O_3$ or $Fe_3O_4$ due to side reactions. These secondary phases tend to be small, sharp peaks scattered among the major reflections and can decrease the electrochemical activity of the cathode.

Along with the onset of new phases, peak widening or loss of intensity can also happen because of reduced crystallite size or greater internal strain in the LFP particles. This is usual in cells that have been cycled numerous times, where the constant expansion and contraction may result in microstructural degradation.

In conclusion, the XRD pattern shows that the aged cathode still maintains its initial LFP crystalline structure, but evidence of degradation is also apparent. These are the existence of amorphous content, possible secondary phases, and minor changes in peak intensity or sharpness. Complete quantitative analysis, like Rietveld refinement, would determine the exact phase composition and measure the extent of degradation more precisely. Comparing this trend with a new LFP reference would provide additional information on how the structure has changed as a result of battery use.

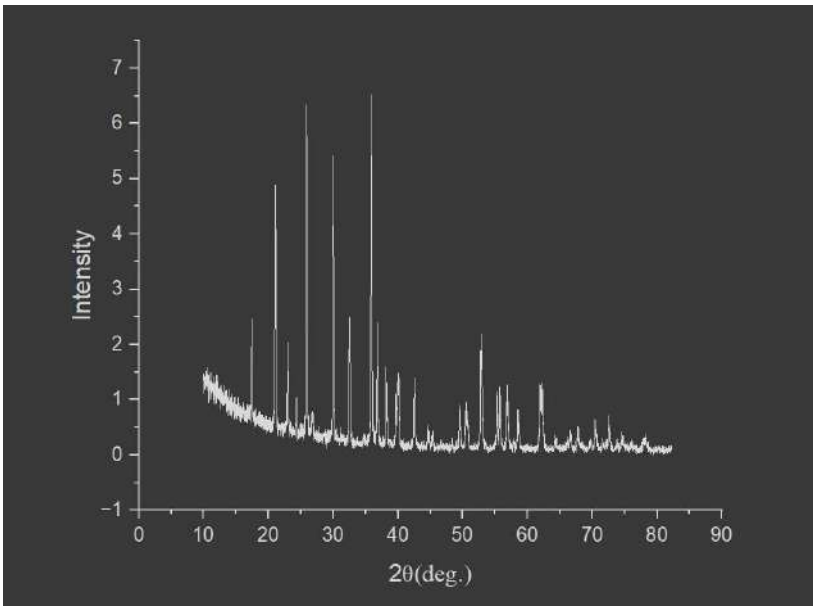



*Figure 7 XRD Graph of Cathode*

## 5.3 XRD Results for Anode

The X-ray diffraction (XRD) pattern offered for the anode material gives important information about the structural features of the material utilized in a lithium-ion battery. From the diffraction peaks, it is evident that the majority phase in the anode is graphitic carbon, which is being extensively utilized as the active material in commercial anodes because of its superior electrochemical characteristics, including good reversibility upon lithium-ion intercalation and deintercalation, good electrical conductivity, and structural stability.

One of the most important characteristics in this pattern is the high and intense peak at $2\theta = 26.5°$ corresponding to the (002) plane of graphite. This plane relates to the layer structure of graphite where carbon atoms are closely packed in two-dimensional sheets with weak van der Waals forces between layers. This (002) peak is a direct measure of graphitization and the intensity and sharpness represent a highly crystalline character. This indicates that the anode material is derived from a good quality graphite source or has received a thermal treatment process to enhance crystallinity.

Further peaks found at around 44.5° and 54.6° corresponding to the (100) and (004) graphite planes respectively provide further evidence for the existence of a highly ordered graphitic structure. These higher-order reflections prove not only high stacking order within the basal planes but also lack of lattice distortion. The sharpness and clarity of these peaks indicate that the material has not incurred extensive structural damage, which typically occurs in aged or heavily cycled battery anodes where lithiation and delithiation mechanisms result in strain and defect creation.

The lack of prominent humps or uneven features in the background of the pattern further suggests a low concentration of amorphous carbon or other disordered phases of carbon,

which could exist in poor-quality or degraded anode materials. This indicates that the sample is devoid of structural defects at a low concentration and with little inactive binder or carbon black which may be introduced to enhance conductivity but whose strong diffraction features are lacking owing to their amorphous nature.

From the degradation point of view, the XRD pattern cannot exhibit any peaks that could be characteristic for byproducts like $Li_2CO_3$, LiOH, or $Li_2O$, which may be present on the anode surface upon cycling, especially as a result of electrolyte decomposition or reactions with minor moisture. The absence of such secondary phases indicates that the sample has either not experienced harsh electrochemical aging or that the SEI (solid electrolyte interphase) layer developed on the anode is predominantly amorphous or undetectable by XRD. Although XRD is not the most sensitive technique for detecting the SEI, its failure to appear in the diffraction pattern supports the notion that the anode structure is still intact.

In short, the XRD pattern of the anode material shows a highly crystalline graphitic phase with intense, sharp peaks that indicate a high-quality and structurally stable anode. This is characteristic of a material that has maintained its performance potential and has not been severely degraded, at least from a crystallographic perspective. For a full evaluation of the anode's electrochemical health, further analyses like Raman spectroscopy, scanning electron microscopy (SEM), or X-ray photoelectron spectroscopy (XPS) would be beneficial, especially for the determination of surface chemistry and SEI formation. However, on the basis of XRD alone, the anode is in fine shape, either as a fresh sample or as one that has survived only modest cycling stress.

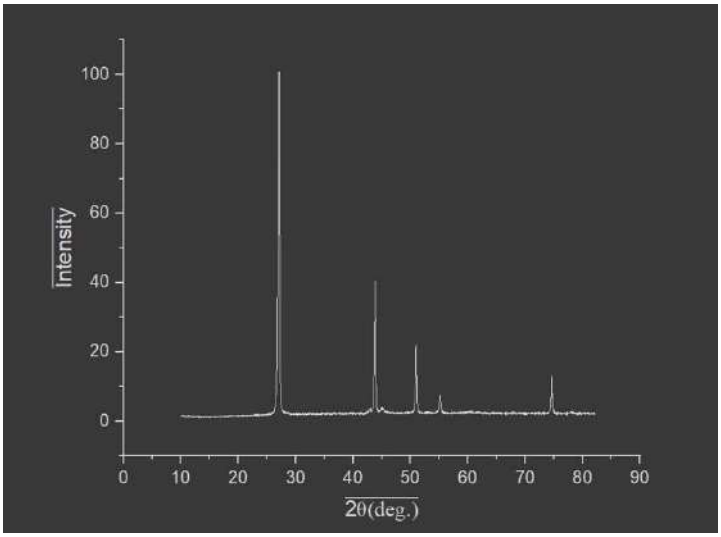



***Figure 8 XRD Graph Of Anode***

# CHAPTER 6: CONCLUTIONS

The in-depth microstructural and crystallographic characterization of the lithium iron phosphate (LFP) battery from an e-rickshaw that is now more than 80% state of health (SoH) declines provides key information regarding the material's present state and viability for second-life usage.

The SEM evaluation of the cathode shows that it has a dense, defect-free microstructure with uniformly spread, submicron-sized particles. These observations are indicative of high sintering quality, good mechanical soundness, and excellent thermal stability, which withstand battery aging without much loss. Likewise, the anode reveals a flaky morphology characteristic of graphitic carbon, with coarse surfaces and with little evidence for structural degradation. Its morphology allows for lithium-ion transport and good electrochemical functionality.

The XRD also verifies that both electrodes have preserved their original crystalline phases. The cathode retains its orthorhombic olivine phase with minimal indications of amorphization and potential secondary phase growth, typical of aged cells. The anode exhibits strong, sharp diffraction peaks typical of highly ordered graphitic carbon, with no observable degradation products or substantial structural deformation.

Together, these results indicate support for the conclusion that although the battery is no longer viable for high-load applications such as e-rickshaws—due to decreased capacity and potential loss of performance under heavy loads—it is structurally and chemically intact. The maintained integrity of both the cathode and anode materials makes the battery a prime candidate for second-life applications such as stationary energy storage, low-power backup systems, or other non-mobility applications.

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